

Making Sense of Psychological Data via Functions and Factors

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Outline of Talk

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Smoothing Spline ANOVA

Regression Background

A **regression model** relates the response variable Y to the independent variables $X_1, \dots, X_p$ using some systematic rule.

For example, the **general linear model (GLM)** assumes that

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i \quad (1)$$

for $i = 1, \dots, n$ where

- y_i is the observed response variable for the i -th subject
- x_{ij} is the j -th observed predictor for the i -th subject
- $(\beta_0, \beta_1, \dots, \beta_p)$ are the unknown regression coefficients
- $\epsilon_i \stackrel{\text{iid}}{\sim} \text{N}(0, \sigma^2)$ is unobserved Gaussian measurement error

Smoothing Spline ANOVA

A **smoothing spline analysis of variance (SSANOVA)** model is a statistical framework for estimating functional relationships.

- See Gu (2013), Gu and Wahba (1991), Helwig and Ma (2015), Kim and Gu (2004), Ma et al. (2015), Wahba (1990)

An SSANOVA model has the form

$$y_i = \eta(x_{i1}, \dots, x_{ip}) + \epsilon_i \quad (2)$$

for $i = 1, \dots, n$ where

- y_i is the observed response variable for the i -th subject
- x_{ij} is the j -th observed predictor for the i -th subject
- η is an unknown function relating Y and $X_1, \dots, X_p$
- $\epsilon_i \stackrel{\text{iid}}{\sim} \text{N}(0, \sigma^2)$ is unobserved Gaussian measurement error

SSANOVA Model Building

With $p > 1$ predictors, we need to specify the form of the function.

For example, with $p = 2$ predictors we have two possible models

additive : $\eta(x_1, x_2) = \eta_0 + \eta_1(x_1) + \eta_2(x_2)$

interactive : $\eta(x_1, x_2) = \eta_0 + \eta_1(x_1) + \eta_2(x_2) + \eta_{12}(x_1, x_2)$

where

- η_0 is an analog of β_0 (unknown regression intercept)
- η_1 is the unknown main effect function for the first predictor
- η_2 is the unknown main effect function for the second predictor
- η_{12} is the unknown interaction effect function

SSANOVA Estimation

To estimate η , we minimize the **penalized least squares** functional

$$\frac{1}{n} \sum_{i=1}^n (y_i - \eta(\mathbf{x}_i))^2 + \lambda J_{\boldsymbol{\theta}}(\eta) \quad (3)$$

where

- $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})'$ is the predictor vector
- $J_{\boldsymbol{\theta}}(\cdot) = \sum_{k=1}^s \theta_s J_s(\cdot)$ is a non-negative **penalty functional**
- λ and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_s)'$ are non-negative **smoothing parameters**

Generalized Cross-Validation criterion (Craven & Wahba, 1979) can be used to estimate the smoothing parameters.

Overview of Self-Esteem Data¹

Online survey data from personal testing website.

- Data collected between 2011 and 2014 (posted Feb-15-2014)
- <http://personality-testing.info/>

Rosenberg Self-Esteem Scale (Rosenberg, 1965).

- 10-item self-report measure of self-esteem
- Rate agreement with items on 4-point Likert scale

Data from $n = 44,488$ subjects (62% female)

¹Joint work with my undergraduate student Mark R. Ruprecht.

Visualization of Self-Esteem Data

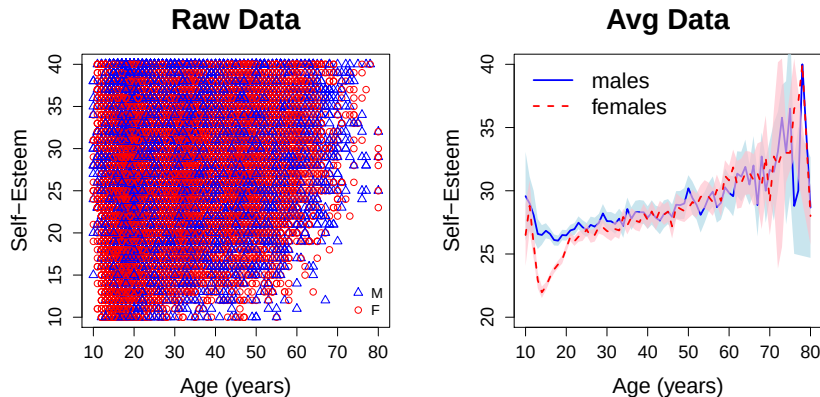


Figure 1: Raw data (left) and average Rosenberg Self-Esteem Scale total score by age and gender (right). Shaded areas are 95% pointwise confidence intervals.

Two-Way SSANOVA Model for Self-Esteem Data

To model self-esteem, we can use a two-way SSANOVA model with

- Cubic smoothing spline for age
- Nominal smoothing spline for gender

additive : $\text{RSES}_i = \eta_0 + \eta_A(\text{age}_i) + \eta_G(\text{gender}_i) + \epsilon_i$

interactive : $\text{RSES}_i = \eta_0 + \eta_A(\text{age}_i) + \eta_G(\text{gender}_i) + \eta_{AG}(\text{age}_i, \text{gender}_i) + \epsilon_i$

Table 1: Fit statistics for two-way (age-gender) SSANOVA models.

Model	R^2	Akaike IC	Bayesian IC
Additive	0.07	824.3	1052.3
Interaction	0.08	484.3	769.1

Note. IC = Information Criterion

SSANOVA Model Predictions for Self-Esteem Data

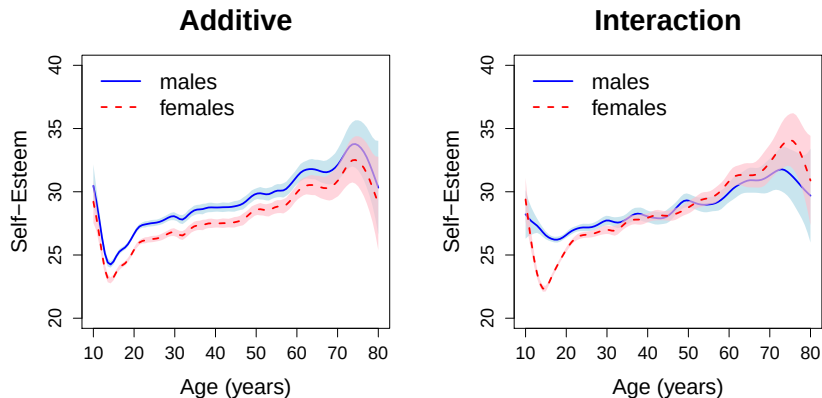


Figure 2: Estimated self-esteem function from additive (left) and interaction (right) two-way SSANOVA model. Shaded areas are 95% Bayesian confidence intervals.

Overview of Twitter Data²

Data from Helwig, Gao, Wang, and Ma (2015).

- Data collected at the CyberGIS Center (University of Illinois)
- <http://cybergis.illinois.edu/>

Have 10 million Tweets collected over the course of one week.

- Each tweet has time stamp (data, hour, sec)
- Each tweet has GPS (latitude, longitude) information

Goal is to understand spatiotemporal trends in the data.

²Joint work with Yizhao Gao & Shaowen Wang (UIUC) and Ping Ma (UGA)

Visualization of Twitter Trends

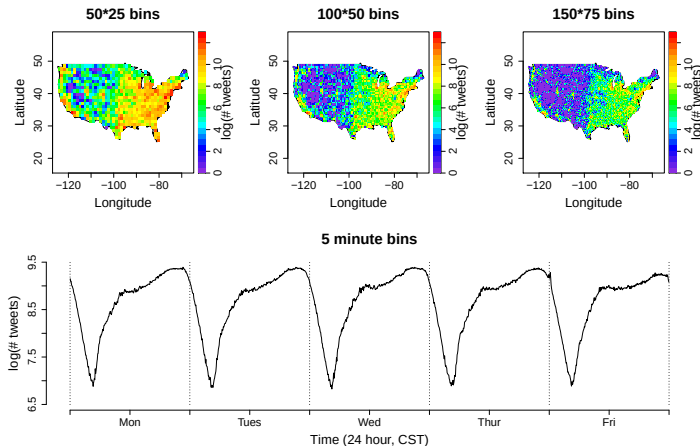


Figure 3: Top: two-dimensional histograms (log scale) using various numbers of bins. Bottom: line plot binning tweets every five minutes. From Helwig et al. (2015)

Visualization of Twitter Time Zone Assignment

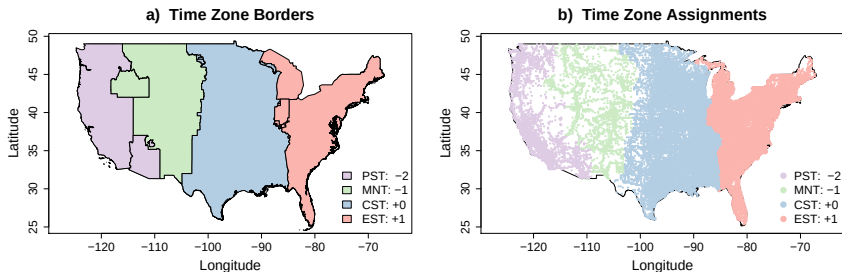


Figure 4: (a) Time zone borders defined by Eric Muller's shape files, along with the Daylight Savings Time offset relative to Central Standard Time. (b) Visualization of tweet time zone assignments. From Helwig et al. (2015)

Two-Way SSANOVA Model for Twitter Data

To model Twitter trends, we can use a two-way SSANOVA model with

- Cubic thin-plate spline for spatial (latitude, longitude) effect
- Periodic cubic smoothing spline for 24-hour time effect

additive : $\log(\# \text{ tweets}_i) = \eta_0 + \eta_S(\text{space}_i) + \eta_T(\text{time}_i) + \epsilon_i$

interactive : $\log(\# \text{ tweets}_i) = \eta_0 + \eta_S(\text{space}_i) + \eta_T(\text{time}_i) + \eta_{ST}(\text{space}_i, \text{time}_i) + \epsilon_i$

Table 2: Fit statistics for two-way (space-time) SSANOVA models.

TZ	Model	R^2	Akaike IC	Bayesian IC
CST	add	0.782	81564.09	92097.07
CST	int	0.797	80428.38	94394.93
Local	add	0.793	80066.68	90648.61
Local	int	0.797	80424.61	94810.51

Note. TZ = Time Zone, IC = Information Criterion

SSANOVA Model Predictions for Twitter Data

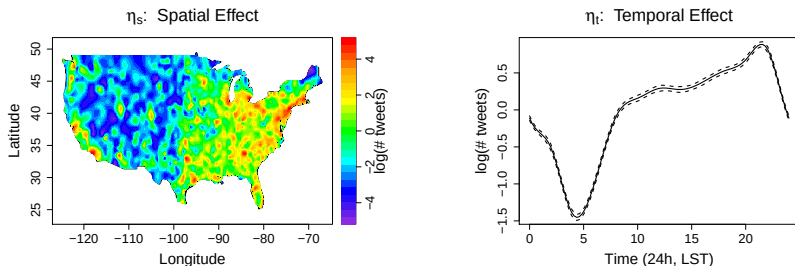


Figure 5: Estimated space (left) and time (right) main effect functions from additive two-way SSANOVA model.

Overview of Smiling Data³

Developed 3D model of face capable of expressing different emotions

- Manipulated spatial and temporal properties of smile
- Created short animations of different expressions

Collected rating data from $n = 808$ subjects at Minnesota State Fair

- 519 females and 289 males; ages 18 to 82
- Subjects rated effectiveness of smile animations

Goal is to understand spatiotemporal properties of an effective smile.

³Joint work with Sofia Lyford-Pike (UMN, MED) & Stephen J. Guy (UMN, CSE)

Visualization of Smile Animations

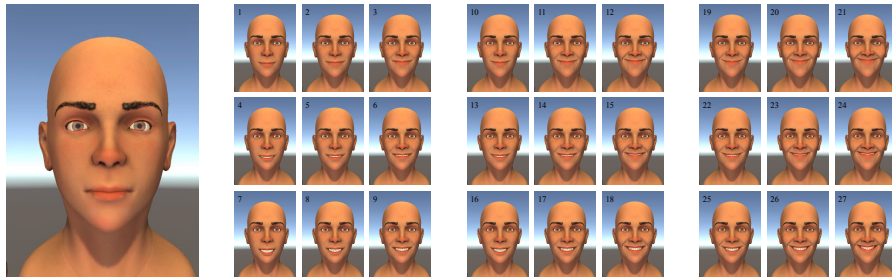


Figure 6: Neutral face used at beginning of all animations (left), and the 27 different smiling faces used in the last frame of the animation. The numbers 1–27 have been included post hoc for labeling purposes and were not present in the animations.

Semiparametric Mixed-Effects Model for Smiling Data

To model the smile effectiveness ratings, we used the model

$$y_{ij} = \beta_0 + \beta_1 t_j + \beta_2 m_i + \eta(a_j, e_j, l_j) + s_i + \epsilon_{ij}$$

where

- y_{ij} is standardized rating of i -th subject on j -th stimuli
- $t_j = 1$ if the j -th stimuli was 350 ms ($t_j = 0$ otherwise)
- $m_i = 1$ if the i -th subject was male ($m_i = 0$ otherwise)
- a_j, e_j, l_j are the *angle*, *extent*, and *lower lip* of j -th smile
- $s_i \stackrel{\text{iid}}{\sim} N(0, \theta^2)$ is a random intercept for the i -th subject
- $\epsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ is error for i -th subject on j -th stimuli

SSANOVA Model Predictions for Smiling Data (part 1)

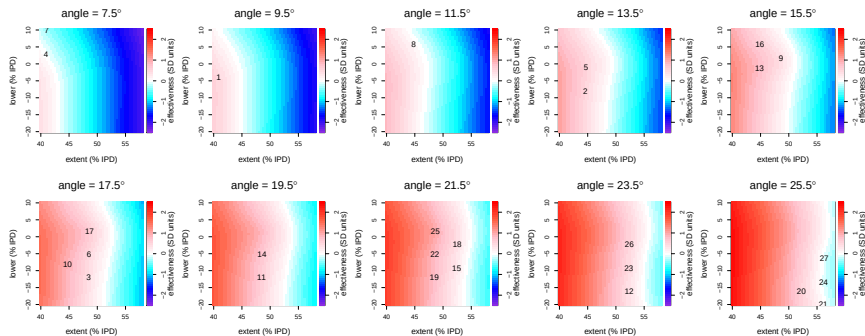


Figure 7: Predicted effectiveness for different angles (a_i), extents (e_j), and lower lips (l_j). The red colors correspond to good (i.e., above neutral) effectiveness ratings, whereas the blue colors correspond to bad (i.e., below neutral) effectiveness ratings. The numbers 1–27 are included in the subplots at the location that best describes the parameters of the 27 stimuli.

SSANOVA Model Predictions for Smiling Data (part 2)

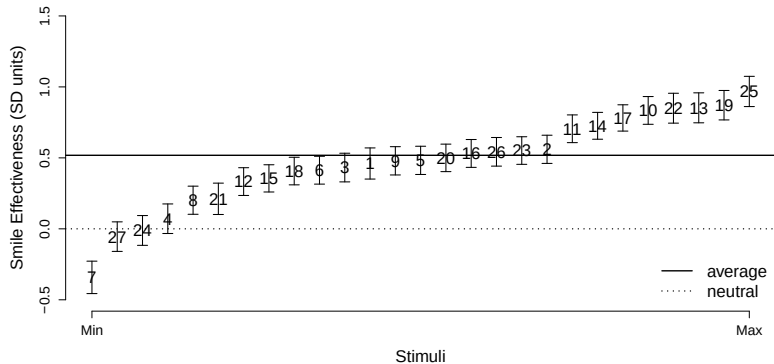


Figure 8: Predicted effectiveness rating with 90% confidence intervals.

Summary of SSANOVA

SSANOVA offers a flexible framework for modeling psychological data.

- Unlike GLM, don't need to specify form of predictors a priori
- Standard statistical model building is possible
- Fixed and/or random effects extensions are possible

Generalized SSANOVA (Wahba et al., 1995) model extends idea.

- Applicable to data from any exponential family
- Binomial, Poisson, Gamma, Negative Binomial, etc.

Some relevant R (R Core Team, 2015) packages:

- `bigsplines` (Helwig, 2015)
- `mgcv` (Wood, 2015)
- `gss` (Gu, 2014)

Multiway Data Analysis

Bilinear Models (PCA, FA, ICA)

Let $\mathbf{X} = \{x_{ij}\}_{I \times J}$ where x_{ij} is i -th occasion of j -th variable.

A **bilinear model** assumes that

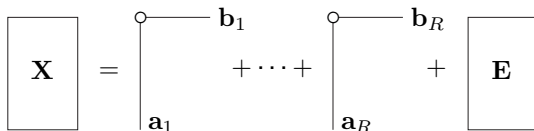
$$\mathbf{X} = \mathbf{AB}' + \mathbf{E} \quad \longleftrightarrow \quad x_{ij} = \sum_{r=1}^R a_{ir}b_{jr} + e_{ij} \quad (4)$$

where

- $\mathbf{A} = \{a_{ir}\}_{I \times R}$ with a_{ir} denoting the weight (**factor score**) of the i -th occasion on the r -th factor/component
- $\mathbf{B} = \{b_{jr}\}_{J \times R}$ with b_{jr} denoting the weight (**factor loading**) of the j -th variable on the r -th factor/component
- $\mathbf{E} = \{e_{ij}\}_{I \times J}$ with e_{ij} denoting the error term of the j -th variable on the i -th occasion

Visualization of Multilinear Structure

$$\text{a) PCA: } \mathbf{X} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r + \mathbf{E}$$



$$\text{b) Parafac: } \underline{\mathbf{X}} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r + \underline{\mathbf{E}}$$

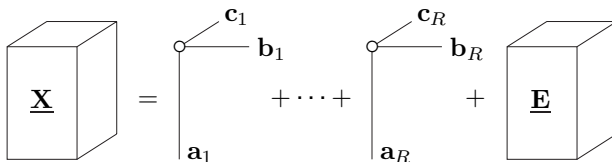


Figure 9: Visualization of multilinear structure from Helwig, Hong, and Polk (2012)

Parafac: Parallel Factor Analysis

$\mathbf{X}_k = \{x_{ij(k)}\}_{I \times J}$ is k -th subject's I occasions by J variables data matrix.

The **Parallel Factors (Parafac)** model (Harshman, 1970) has the form

$$\mathbf{X}_k = \mathbf{A}\mathbf{C}_k\mathbf{B}' + \mathbf{E}_k \quad \longleftrightarrow \quad x_{ijk} = \sum_{r=1}^R a_{ir}b_{jr}c_{kr} + e_{ijk} \quad (5)$$

where

- $\mathbf{C}_k = \text{diag}(c_{k1}, \dots, c_{kR})$ with c_{kr} denoting the weight (**factor score**) of the k -th subject on the r -th factor/component
- $\mathbf{E}_k$ is k -th subject's I occasions by J variables error matrix

Individual Differences Scaling (INDSCAL; Carroll & Chang, 1970) is the equivalent of Parafac developed in the MDS context.

Parafac2: Parallel Factor Analysis-2

$\mathbf{X}_k = \{x_{ij(k)}\}_{I_k \times J}$ is k -th subject's I_k occasions by J variables data matrix.

The **Parafac2** model (Harshman, 1972) has the form

$$\mathbf{X}_k = \mathbf{A}_k \mathbf{C}_k \mathbf{B}' + \mathbf{E}_k \quad \text{subject to } \mathbf{A}_k' \mathbf{A}_k = \mathbf{\Phi} \quad (6)$$

where

- $\mathbf{A}_k = \{a_{ir(k)}\}_{I_k \times R}$ with $a_{ir(k)}$ denoting the k -th subject's weight of the i -th occasion on the r -th factor/component
- $\mathbf{\Phi}$ is the common crossproduct matrix for all $k = 1, \dots, K$

One of four different **Simultaneous Component Analysis (SCA)** models (see Helwig, 2013; Timmerman & Kiers, 2003).

Parafac and Parafac2 Parameter Estimation

Many approaches exist for estimating parameters of multiway models (see Kolda & Bader, 2009; Kroonenberg, 2008; Smilde et al., 2004).

Alternating least squares (ALS) finds the weights that minimize SSE.

Can impose various constraints in the ALS algorithm

- Orthogonality constraints (Harshman & Lundy, 1994)
- Non-negativity constraints (Bro & De Jong, 1997)
- Smoothness constraints (Timmerman & Kiers, 2002)
- Structural constraints (Helwig, in prep)

Overview of Schematic Faces Data

Dissimilarity data from Köhn (2006).

- Data collected at the University of Illinois years ago.

$J = 12$ schematic faces that vary according three dimensions:

- Face shape (circle vs oval)
- Eyes (open vs solid)
- Mouth (frown, flat, smile)

$K = 22$ students rated dissimilarity between all $66 = J(J - 1)/2$ pairs

Visualization of Schematic Faces

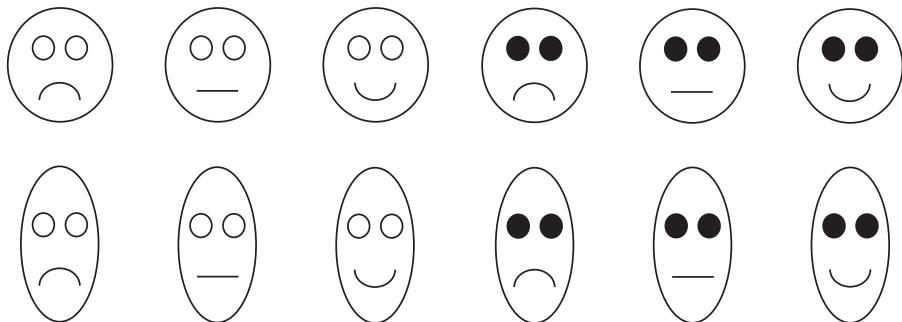


Figure 10: Schematic faces stimuli that vary according to three dimensions: face shape (circle, oval), eyes (open, solid), and mouth (frown, flat, smile).

INDSCAL Results (stimuli weights)

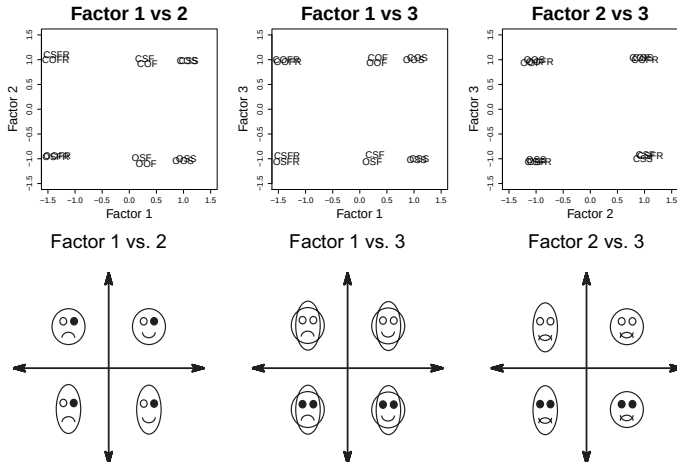


Figure 11: Estimated INDSCAL weights for $J = 12$ schematic faces.

INDSCAL Results (subject weights)

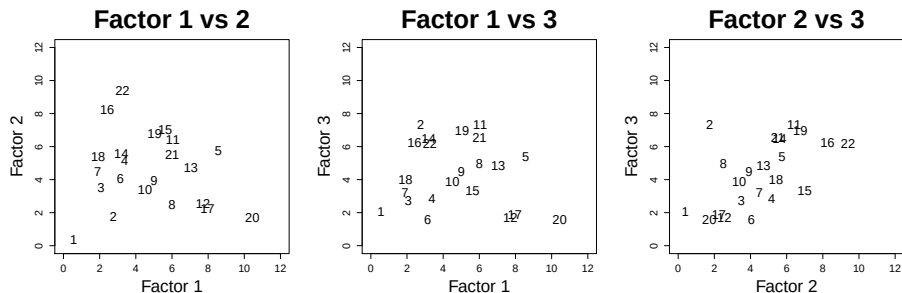


Figure 12: Estimated INDSCAL weights for $K = 22$ subjects.

Overview of Color-Emotion Rating Data⁴

$I = 37$ different colors used as stimuli

- Colors varied according to hue, saturation, and lightness
- All colors were presented against neutral gray background

Colors were rated on $J = 9$ bidimensional features/emotions

Agitated vs Calm	Disgusting vs Appealing	Sad vs Happy
Bland vs Spicy	Dissonant vs Harmonious	Serious vs Whimsical
Cool vs Warm	Quiet vs Loud	Simple vs Complex

$K = 30$ subjects rated each color on each feature (from -200 to 200).

⁴Joint work with Kelly L. Whiteford (UMN, PhD student), Karen B. Schloss (Brown), and Stephen E. Palmer (UC-Berkeley)

Visualization of Color Stimuli

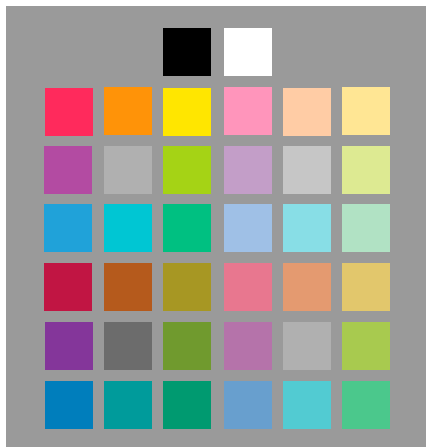


Figure 13: Colors stimuli from Palmer, Schloss, Xu, and Prado-León (2013).

Parafac Results (stimuli & features weights)

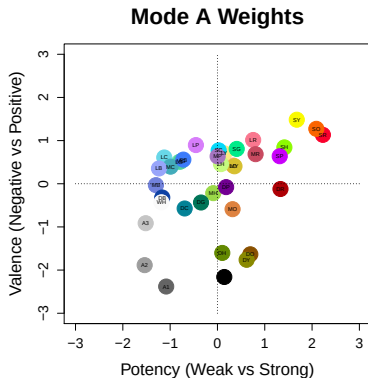


Table 3: Parafac Mode B weights.

Feature (Low.High)	b_{j1}	b_{j2}
Agitated.Calm	− 1.43	0.79
Bland.Spicy	1.27	0.35
Cool.Warm	1.32	0.35
Disgusting.Appealing	−0.56	1.50
Dissonant.Harmonious	−0.8	1.34
Quiet.Loud	1.28	0.32
Sad.Happy	0.56	1.25
Serious.Whimsical	0.20	1.56
Simple.Complex	0.81	−0.11

Note: Weights with magnitudes greater than one are boldfaced.

Figure 14: Parafac Mode A weights.

Overview of Alcohol Consumption Data

Data from National Institute on Alcohol Abuse and Alcoholism

- Measuring alcohol consumption in US from 1970–2013
- $I = 44$ years of data from alcohol sales records

Using $J = 12$ observed variables to measure $R = 3$ factors

- 3 factors: beer, spirits, and wine consumption
- 4 indicators per factor: (vol bev, vol alc) $\times$ (age 14+, age 21+)

$K = 51$ locations (50 US states and District of Columbia).

Visualization of Alcohol Consumption by State

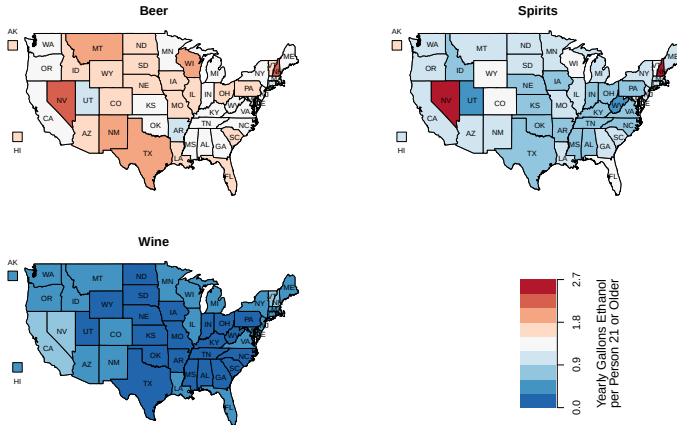


Figure 15: Average alcohol consumption across time (1970-2013) within state.

Parafac and Parafac2 Time Weights

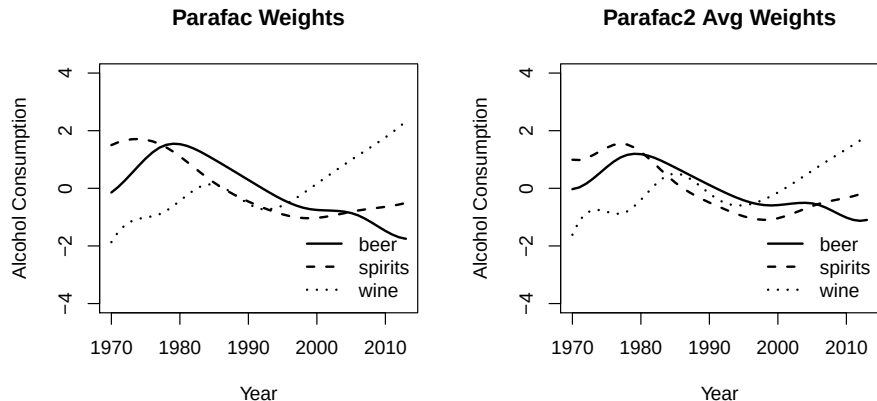


Figure 16: Estimated Parafac and Parafac2 time weights.

Parafac2 Implied Alcohol Consumption by State

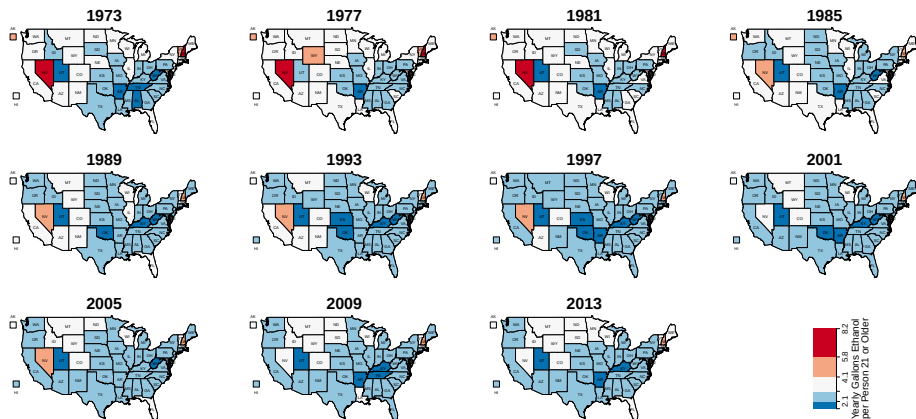


Figure 17: Parafac2 estimated total alcohol consumption by state for select years.

Summary of Multiway Data Analysis

Multiway models are extensions of classic bilinear models.

- Appropriate for analyzing a collection of data matrices/arrays
- Can model group and individual differences in multivariate data

Models can be extended/customized in a variety of ways.

- Incorporating constraints and/or fixing weights
- Hybrid models that combine two multiway models

Some relevant R (R Core Team, 2015) packages:

- `multiway` (Helwig, 2016)
- `PTAk` (Leibovici, 2010)
- `ThreeWay` (Giordani et al., 2014)

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